Build: Pagination Component

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Problem: Build a pagination component with page numbers, previous/next buttons, and dynamic data display.

What This Tests: Pagination math, derived state, controlled current page, edge case handling.

jsx

JSX Copy

```
function Pagination({ totalItems, itemsPerPage = 10, currentPage, onPageCh
  const totalPages = Math.ceil(totalItems / itemsPerPage);

  const getPageNumbers = () => {
    const pages = [];
    const start = Math.max(1, currentPage - 2);
    const end = Math.min(totalPages, currentPage + 2);

    if (start > 1) { pages.push(1); if (start > 2) pages.push('...'); }
    for (let i = start; i ≤ end; i++) pages.push(i);
    if (end < totalPages) { if (end < totalPages - 1) pages.push('...'); p

    return pages;
  };

  return (
    <div style={{ display: 'flex', gap: 4, alignItems: 'center' }}>
      <button onClick={() => onPageChange(currentPage - 1)} disabled={curr

      {getPageNumbers().map((page, i) =>
        page === '...' ? (
          <span key={`dots-${i}`}>...</span>
```

```
            </button>
          )
        )}

        <button onClick={() => onPageChange(currentPage + 1)} disabled={curr

        <span style={{ marginLeft: 8, color: '#666' }}>Page {currentPage} of
      </div>
    );
}

// Usage with data
function UserTable() {
  const [page, setPage] = useState(1);
  const allUsers = Array.from({ length: 95 }, (_, i) => ({ id: i + 1, name
  const perPage = 10;

  const paginatedUsers = allUsers.slice((page - 1) * perPage, page * perPa

  return (
    <div>
      <ul>{paginatedUsers.map(u => <li key={u.id}>{u.name}</li>)}</ul>
      <Pagination totalItems={allUsers.length} itemsPerPage={perPage} curr
    </div>
  );
}
```

Key Concepts: Pagination math (totalPages, slice calculation), sliding window for page numbers with ellipsis, disabled prev/next at boundaries, active page highlighting, reusable controlled component

QUICK CHECK — no marks, just for you.

1. In the UserTable usage, 95 users are rendered 10 per page. Using the Pagination component's own formula, what is totalPages?

- ○ 9.5

- ◯ 11
- ◯ 9

Check answer

2. UserTable slices its data with allUsers.slice((page - 1) * perPage, page * perPage) where perPage is 10. When page is 3, which users are displayed?

- ◯ Users 21 through 30
- ◯ Users 30 through 39
- ◯ Users 20 through 29
- ◯ Users 21 through 31

Check answer

3. getPageNumbers builds a sliding window with start = Math.max(1, currentPage - 2) and end = Math.min(totalPages, currentPage + 2), plus first/last page and ellipsis handling. With totalPages = 10 and currentPage = 5, what array does it return?

- ◯ [1, '...', 4, 5, 6, '...', 10]
- ◯ [1, 2, '...', 5, '...', 9, 10]
- ◯ [1, '...', 3, 4, 5, 6, 7, '...', 10]
- ◯ [3, 4, 5, 6, 7]

Check answer

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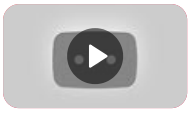
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- Make pagination in React
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- Pagination without any libraries
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Hand-picked references for this lesson.